

AOW Technical – Market Model

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What is it?

The Market Model is a prototype for a web based application. It allows us to use the capabilities of the web application to test various market based reports to support ISK making activities in Eve Online.

It has two parts:

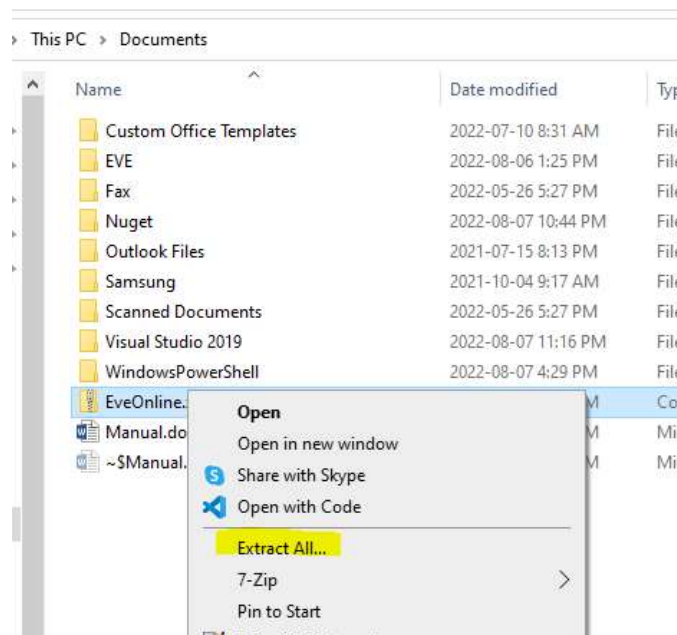
- Data Collection – two tools are provided to collect data from the Eve Swagger API. One tool collects non-personal data (public) and does not require an Eve login. The second tool collects personalized information (private), and hence does require a login to Eve.
- Data Visualization – two spreadsheets are provided. One is for Microsoft Excel, the other can be used by Apache OpenOffice. Both spreadsheets are configured with identical macro-based functions which can be used to extract and use the data collected by the first too.

Installation of the market model

The model can be installed to any folder you wish. Depending on where you “install” it, Powershell may require reconfiguration to allow you to run scripts from that location.

In our example, we will use the \Documents\ folder to make things easier.

Copy the ZIP file to the Documents folder using Windows Explorer. Right-click on the ZIP file and select “Extract All...”. Click the “Extract” or “Next” button in the dialog that appears.



You should have a folder under \Documents named “EveOnline” with all the scripts, data and spreadsheets in it. Delete the ZIP file since it is no longer needed.

Installation of Open Office

Open Office is an open source suite of applications similar to Microsoft Office and is available from the Apache Software Foundation. Unlike Microsoft Office, it is free for personal use.

Before you install it, you need a 32 bit version of Java (even if you are running a 64 bit version of Windows).

The official download page for Java is here: [Java Downloads for All Operating Systems](#)

 Windows  Which download should I choose?			
	Windows Online filesize: 2.16 MB	Instructions	After installing Java, you may need to restart your browser in order to enable Java in your browser.
	Windows Offline filesize: 72.73 MB	Instructions	
	Windows Offline (64-bit) filesize: 83.46 MB	Instructions	
If you use 32-bit and 64-bit browsers interchangeably, you will need to install both 32-bit and 64-bit Java in order to have the Java plug-in for both browsers. » FAQ about 64-bit Java for Windows			

Click the highlighted link to download the installer to your \Downloads folder and run the file once it has downloaded. Accept all the defaults until the installer has finished.

The official download page for Open Office is here: [Apache OpenOffice - Official Download](#)

home » download

Download Apache OpenOffice
(Hosted by SourceForge.net - A trusted website)

Select your favorite operating system, language and version:

Windows 32-bit (x86) (EXE) English [US] 4.1.13

Download full installation Download language pack

[Important hints for Windows 32-bit \(x86\) \(EXE\)](#)

Release: Milestone AOO4113m1 | Build ID 9810 | Git hash 281f0d3533 | Released 2022-07-22 | [Release Notes](#)
Full installation: File size ~ 134 MByte | Signatures and hashes: [KEYS](#), [ASC](#), [SHA256](#), [SHA512](#)
Language pack: File size ~ 18 MByte | Signatures and hashes: [KEYS](#), [ASC](#), [SHA256](#), [SHA512](#)

[What is a language pack?](#) [How to verify the download?](#) [Report broken link](#)

Click the highlighted link to download the installer to your \Downloads folder and run the file once it has downloaded. Accept all the defaults until the installer has finished.

Security Warning! READ THIS!

The spreadsheets rely heavily on macros. We have not signed them so in order for the spreadsheets to work, you must enable all macros.

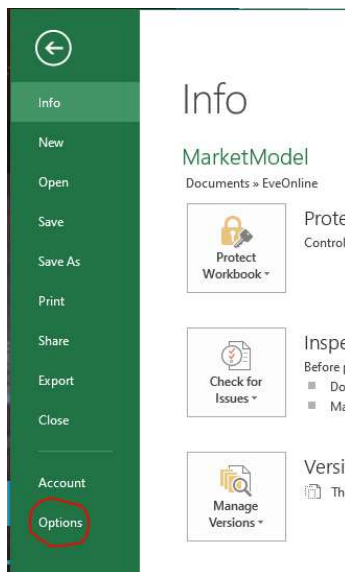
This is risky. For instance, if you receive a file from someone with embedded macros and you enable all macros, it is possible that opening that file you received could infect you with malware. **NEVER open a**

file from someone you do not know, or got from a source you do not trust with settings that enable all macros.

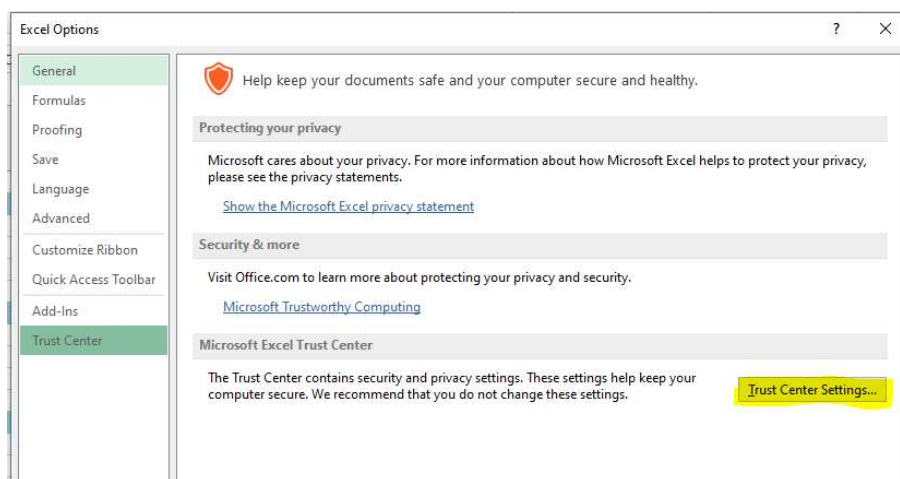
You may choose to enable macros using the following instructions before you open the model, and when finished, reverse the change to ensure you are secure.

Configuration of Excel for macros

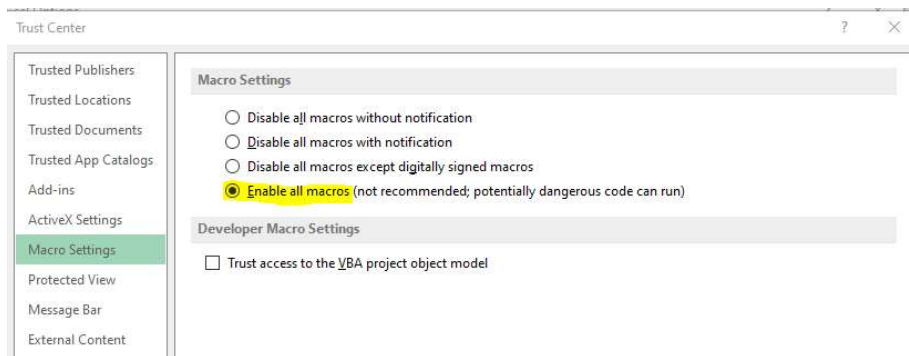
Click on “File” on the menu bar. Click on “Options.”



In the dialog that appears, select “Trust Center” in the left pane and then click the “Trust Center Settings” button.



Click on “Macro Settings” in the left pane. Note the settings so you can put them back if security is a concern. Enable all Macros by clicking the highlighted option.

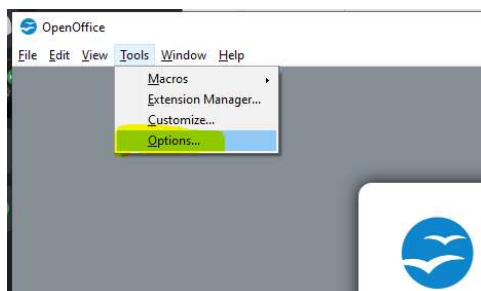


Configuration of Open Office for macros

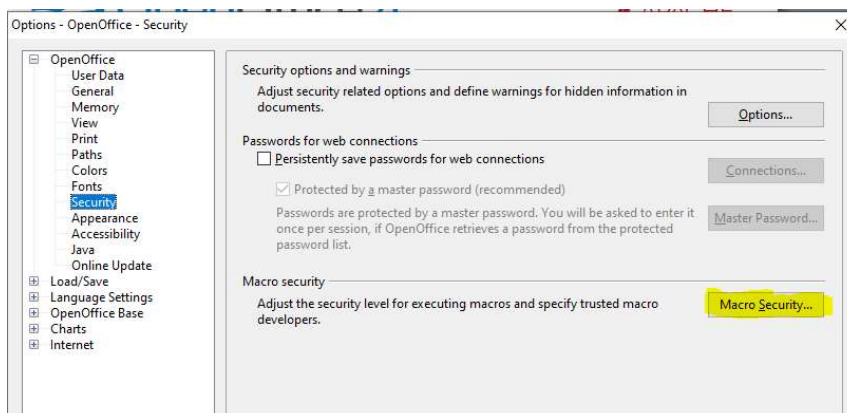
You should have an OpenOffice icon on your desktop, or in your Start Menu. Use it to launch OpenOffice.



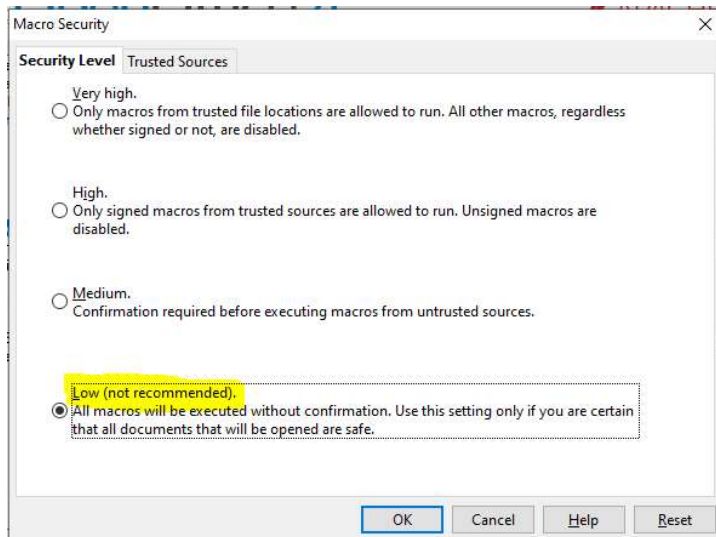
From the menu, select “Tools”, “Options.”



In the left pane, click “OpenOffice”, “Security” and click the “Macro Security” button.



Note the settings so you can put them back if security is a concern. Enable all Macros by clicking the highlighted option.



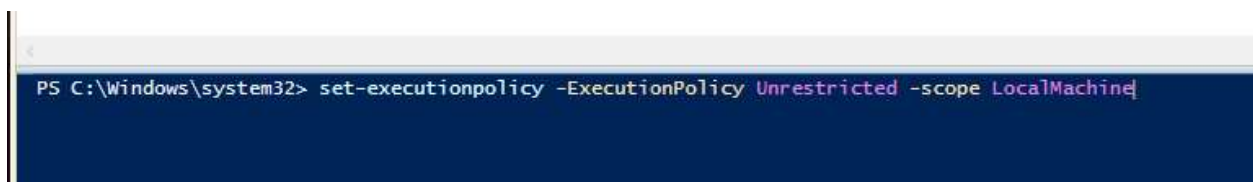
Data Collection

The data collection scripts are written in Windows Powershell, a very powerful scripting language that is installed on every Windows computer since Windows 7. This means that no installation is necessary in order to use the scripts.

There are several security settings that may need to be enabled to allow the script(s) to run. You should only have to change them the first time you run the scripts. In order to change the security settings in Powershell, you must run it as administrator.

You can find Powershell in the Start Menu under “Windows Powershell”, locate the “Windows Powershell ISE” entry, right-click on it and choose “More”, then “Run as Administrator.”

In the blue window (the console), type “set-executionpolicy -ExecutionPolicy Unrestricted -scope LocalMachine” and hit Enter.



In the dialog that pops up, click “Yes to All” and close the Powershell ISE.

** Edits pending here, file may need to be unblocked.”

Input Files

The script saves its data on Comma Separated Value (.csv) files. This makes them very easy to import to a spreadsheet or a database.

There are two mandatory files that control the behaviour of the script:

WatchMarkets.csv – this file specifies the regions and markets to save the pricing and order data from. It is possible to add or remove entries and this either increases or reduces the overall script run time and the amount of data generated.

Columns are (in order):

1. Market ID – this is a numerical identifier for the station or structure that hosts the market. The ID can usually be found in one of two other CSV files: Structures.csv (player stations) or Stations.csv (NPC stations.) If this data is missing, the script will attempt to locate the market by the name and add the Market ID to the file upon completion.
2. Name – this is a short version for the market name and is used by the spreadsheet for data lookups. For instance “Jita” for “Jita IV - Moon 4 - Caldari Navy Assembly Plant.”
3. Station Name – this is the long version of the market name, “Jita IV - Moon 4 - Caldari Navy Assembly Plant.”
4. Region Name – this is the name of the region and is used for display in the progress bar when data is being downloaded, “The Forge.”
5. Region ID – this is the numerical identifier for the region, it can be found in the Regions.csv file.

WatchItems.csv – this file specifies the item prices and orders you wish to track. It is possible to add or remove entries and this either increases or reduces the overall script run time and the amount of data generated.

Columns are (in order):

1. Item ID – this is a numerical identifier for the product. It can be found in the Types.csv file. For instance, Plagioclase has an item id of 18.
2. Item Name – this is the full name of the product, for example: “Plagioclase.”

The following files are optional. If they do not exist, the script will query Eve for a full update, saving the data for future use and updates. If they exist, they will be read first and any changes that are found (new stations, items, etc.) will be made and saved over time.

Regions.csv – this is a list of regions with their ID’s.

Columns are (in order):

1. Region ID – the numerical identifier for the region, for example: 10000001
2. Region Name – the full name of the region, for example: Derelik

Stations.csv – this is a list of every NPC station in Eve with its associated numerical identifier and the identifier for the system it is located in.

Columns are (in order):

1. Station ID – the numerical identifier for the station, for example: 60003760
2. Station Name – the full name of the station, for example: Jita IV - Moon 4 - Caldari Navy Assembly Plant
3. System ID – the numerical identifier for the system the station is in, for example: 30000142

Structures.csv – this is a list of every “open” player structure, or if updated, every structure you have access to with its associated numerical identifier and the identifier for the system it is located in.

Columns are (in order):

1. Structure ID – the numerical identifier for the structure, for example: 1036974821465
2. Structure Name – the full name of the structure, for example: Jel - Jel clone center
3. System ID – the numerical identifier for the system the structure is in, for example: 30005222

Systems.csv – this is a list of every system in Eve with its numerical ID, name and security level.

Columns are (in order):

1. System ID – the numerical identifier for the system, for example: 30005222
2. System Name – the full name of the system, for example: Serren
3. Security – the numerical identifier for the system security, for example: 0.4588785171508789

Types.csv – every item in Eve has a name and an ID, this file contains all known items and their ID’s.

Columns are (in order):

1. Type ID – the numerical identifier for the item, for example: 18
2. Type Name – the full name of the item/type, for example: Plagioclase

Output Files

The script produces two files with new content, the content from previous runs are lost.

Orders.csv – this is a list of every observed order at the markets listed in WatchMarkets.csv, and only for the items listed in WatchItems.csv.

Columns are (in order):

1. Type ID – the numerical identifier for the item, for example: 18
2. Station ID – the station ID for where the order is at, for example: 60003760
3. Transaction Type – Buy or Sell
4. Price – the buy or sell price in ISK, each
5. Total – the total order volume in items
6. Remaining – the remaining number of items for buy or sale

Prices.csv – this is a list of statistical observations based on the prices data.

Columns are (in order):

1. Type ID – the numerical identifier for the item, for example: 18
2. Station ID – the station ID for where the order is at, for example: 60003760
3. Transaction Type – Buy or Sell
4. High – the highest price observed for the item/transaction type in ISK
5. Low – the lowest price observed for the item/transaction type in ISK
6. Average – the average price per unit observed for the item/transaction type in ISK

Collecting the Data

There are two types of data available in the Eve API – public and private. All information that is available can be browsed here: [EVE Swagger Interface \(evetech.net\)](https://eveapi.com/)

As you browse the data, you will see that some data has a “lock” icon and some data does not. This “lock” icon indicates that the data is not public and you have to log in to Eve to retrieve it.

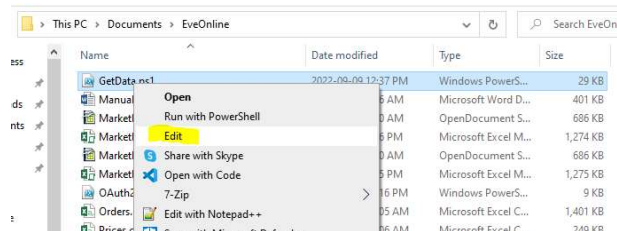


The non-public data is personalized and different for each player.

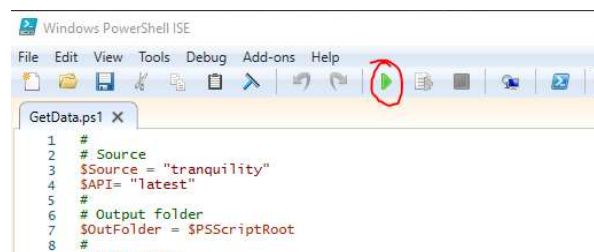
Public Data Refresh (GetData.ps1)

In most cases, you only need to update the public data. The scripts are separated by the type of data they collect to allow collection of data without the need to log in which can be improved with a scheduled and automated data refresh. At this time, all refreshes are manual.

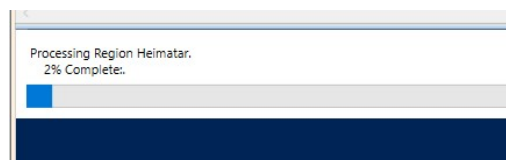
To run the script, open the EveOnline folder in Windows Explorer, select the GetData.ps1 file. Right click on it and choose “Edit.” This will open the script in the Powershell ISE.



The little green “arrow” on the ribbon bar is the “Run” button. Click it.



As the script executes, a progress bar will be displayed as the order data is downloaded for each configured regional market.



Once the script completes, the blue area (console) will look like this, you can close the Powershell ISE like you would any other Windows application. The two date/time entries indicate when the script started and finished.

```
September 18, 2022 6:59:24 AM
September 18, 2022 7:25:56 AM

PS C:\Users\IanWBaxter\Documents\EveOnline>
```

The updated Price and order data is now ready to be updated in whichever data visualization tool you have chosen (Excel or OpenOffice Calc.)

Private Data Refresh (OAuth2.ps1)

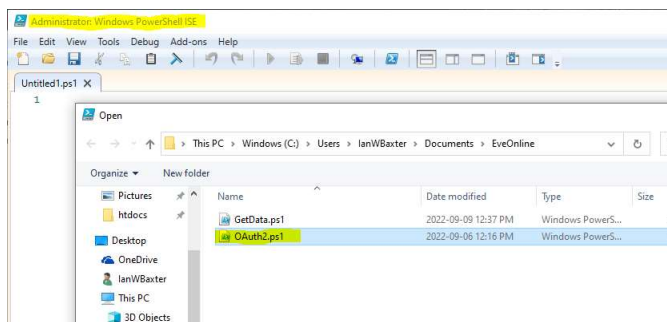
In order to access character-specific data in Eve such as the Structures data, you must provide access to the requested data by signing in to Eve. This is called “Single Sign On” and the process is documented here: [Single Sign On \(SSO\) – EVE Online](#)

Essentially, the script creates a secure, random key, opens your web browser, sends you to the Eve sign on page and passes the sign on page the secure key. Once you have signed in, the Eve sign on page will redirect you to a web server running on your own computer with a verification code that can be used to ensure that nobody has “spoofed” your credentials. The process of creating and running a web server on your computer requires that you have administrative rights so a different process is required to run this script.

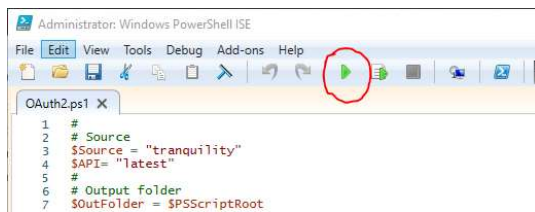
At no time is your Eve Online account information captured or stored by the script, it is impossible to do so. This is also the same method most websites use to allow sign-in via your Facebook, Google, Microsoft or other accounts. Read the link above to understand how to verify that you are using a valid login page for Eve online. This same technique can be used for other single sign on providers.

You can find Powershell in the Start Menu under “Windows Powershell”, locate the “Windows Powershell ISE” entry, right-click on it and choose “More”, then “Run as Administrator.”

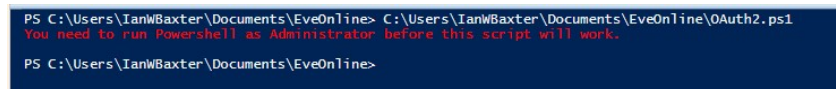
If you did it correctly and approved the application to run as an administrator, Powershell ISE will open. Note that the title will show as “Administrator: Windows Powershell ISE”. Click “File”, “Open” and select the OAuth2.ps1” script.



The little green “arrow” on the ribbon bar is the “Run” button. Click it.

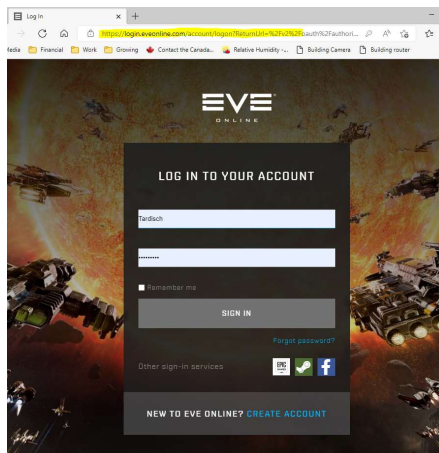


If you did not launch Powershell ISE as an Administrator, the following message will be displayed in the console:

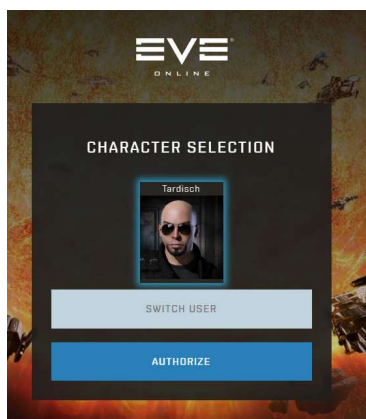


If you have a web server running on your computer, you will see a lot of errors in the Powershell console. You need to stop the web server service for it to be able to create its own web server.

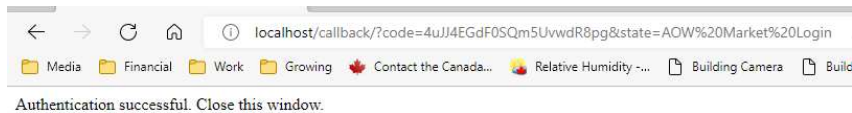
If everything worked correctly, you should see the Eve Online sign in page with no errors in the Powershell console.



Log in to Eve and then select the character whose structure data you wish to update. Click the “Authorize” button.



Your web browser will now be redirected to the temporary web server running on your computer with the authorization code. You can close this window.



The progress bar will display the script's progress. When the progress bar clears, the script has finished.



Data Visualization

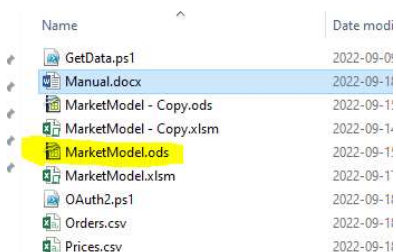
Now that we have the market data, it is time to do something with it. Several possibilities for uses are created as example worksheets in the Excel and OpenOffice models. They are almost identical except for one formula and one automated process which proved too difficult to translate from Excel.

Updating the Prices

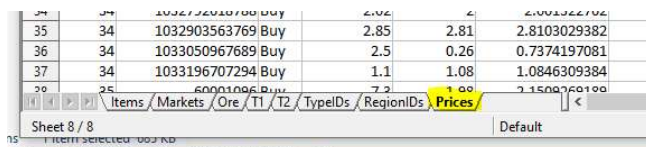
Loading the pricing data into the model is the automation process which could not be made identical between the two model versions.

Open Office

First, open the model – using Windows Explorer, open the Documents\EveOnline folder, locate the OpenOffice file 'Marketmodel.ods' by name, or if you do not see the file extension, then by name and the icon. Use the snapshot below for reference. Double-click on it to open it.



At the bottom of the model window are a bunch of tabs, these are worksheets. Click on the tab for the Prices worksheet.



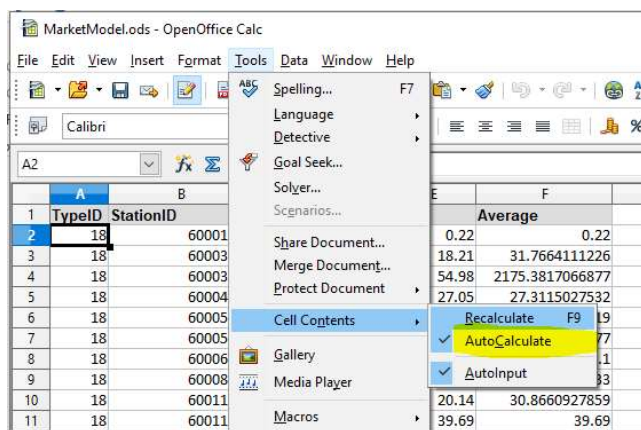
Make sure you are at the top of the sheet data by using the scrollbar at the right of the window. Note that above the sheet data are letters indicating the column name, and numbers at the left to indicate the row number. The two in combination, “A2” give the “address” of the box, or cell.

Click in cell “A2”, in the snapshot below:

	A	B	C	D	E	F
1	TypeID	StationID	Type	High	Low	Average
2	18	60001096	Buy	0.22	0.22	0.22
3	18	60003760	Buy	36.46	18.21	31.7664111226
4	18	60003760	Sell	35600	54.98	2175.3817066877
5	18	60004588	Buy	28.04	27.05	27.3115027532
6	18	60005686	Buy	28.12	0.06	6.968847619
7	18	60005686	Sell	34.98	30	34.9761764577

Before we remove that price data and replace it, we need to make sure that the model will not try to update while we are doing the work. If we do not, then it will take longer to update the data.

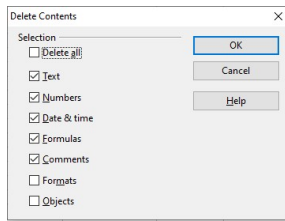
From the menu, select “Tools”, “Cell Contents” and click on “AutoCalculate” to remove the check mark to the left, indicating automatic calculation has been turned off.



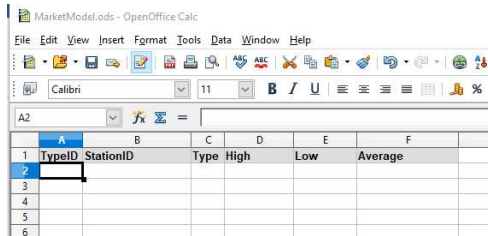
Select the data only by selecting cell “A2” and then hitting the key combination Shift-Ctrl-End. The display will change to show you have selected to the end of the data. Note the highlighted addresses in the snapshot. Two addresses with a colon between them specifies an area, or “range.”

	A	B	C	D	E	F
2642	62530	60003760	Buy	47.01	25.24	40.9937146703
2643	62530	60003760	Sell	99.99	49.93	83.9840200898
2644	62530	60004588	Buy	30.03	30.03	30.03
2645	62530	60005686	Buy	27.27	26.88	27.1338866575
2646	62530	60005686	Sell	46	46	46
2647	62530	60008494	Buy	40	2.1	8.5397309582
2648	62530	60008494	Sell	60	58.73	58.7314298057
2649	62530	60011866	Buy	31.69	30.03	31.3969740677
2650	62530	60011866	Sell	48.3	46.48	47.3189696668
2651	62530	1032792618788	Buy	15.01	15.01	15.01
2652						
2653						

Hit the “Delete” key and make sure you have selected the options shown below, click “OK.”

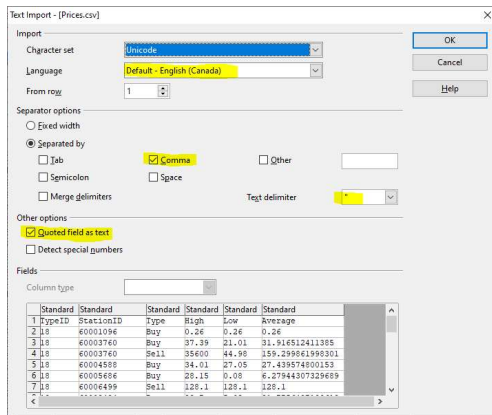


Now hit the key combination Ctrl-Home to go back to the top of the sheet. Select cell “A2” again.

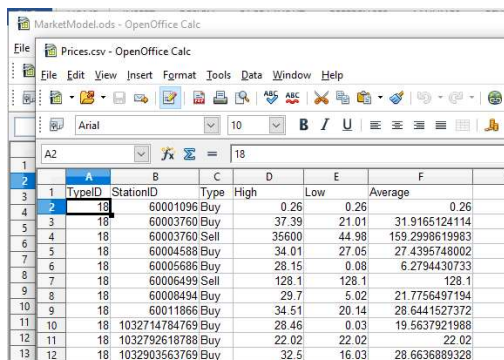


Using Windows Explorer, locate the Prices.csv file (your icon will appear different) and double-click on it to open it if OpenOffice Calc.

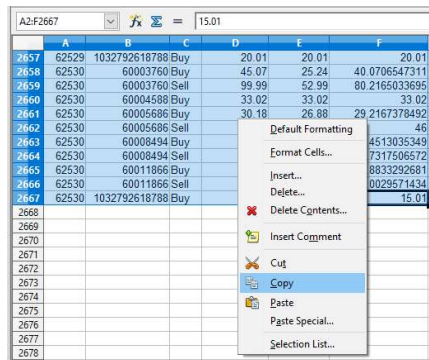
A dialog box will appear. Make sure that the Language is set correctly for your location, the comma is set as a separator, the text delimiter is double quotes and the quoted field as text option is selected. Click “OK.”



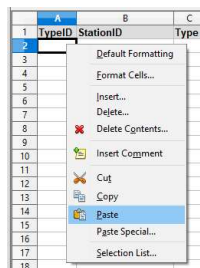
Now that you have the Prices.csv file open, select Cell “A2” from that sheet.



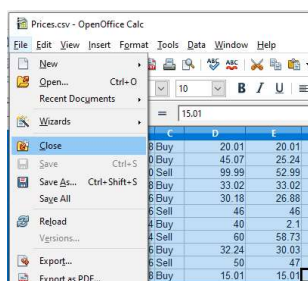
Press Shift-Ctrl-End to select all the data. This time, right-click on the blue area and select “Copy” from the menu that appears.



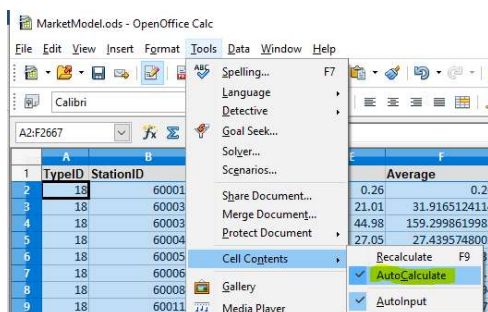
Switch to the “MarketModel.ods - OpenOffice Calc” window, select cell “A2” and right click on it. Select “Paste” from the menu that appears.



Switch to the “Prices.csv – OpenOffice Calc” window and close it.

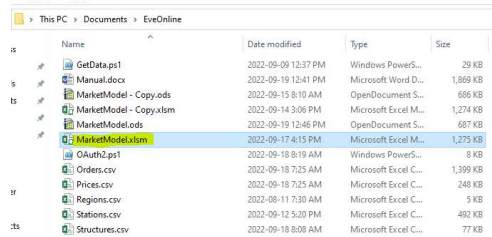


Switch to the “MarketModel.ods - OpenOffice Calc” window. From the menu, select “Tools”, “Cell Contents” and click on “AutoCalculate” to force the formulas in the model to update.



Excel

First, open the model – using Windows Explorer, open the Documents\EveOnline folder, locate the OpenOffice file ‘Marketmodel.xlsm’ by name, or if you do not see the file extension, then by name and the icon. Use the snapshot below for reference. Double-click on it to open it.



At the bottom of the model window are a bunch of tabs, these are worksheets. Click on the tab for the Prices worksheet.

A screenshot of the 'Prices' worksheet in Excel. The table shows market data for various items. The columns are: Item, Market, Watches, Ore, Salvage, Planet, T1, T2, RegionIDs, TypeIDs, and Prices. The data includes rows for items like 60011866 (Buy/Sell) and 1032714794769 (Buy).

Item	Market	Watches	Ore	Salvage	Planet	T1	T2	RegionIDs	TypeIDs	Prices
60011866	Buy		3.39		2.12	3.01				
60011866	Sell		333.00		4.25	5.72				
60015131	Buy		0.10		0.10	0.10				
1032714794769	Buy		2.42		0.30	1.20				
10327175132181	Buy		0.11		0.11	0.11				

Make sure you are at the top of the sheet data by using the scrollbar at the right of the window. Note that above the sheet data are letters indicating the column name, and numbers at the left to indicate the row number. The two in combination, “A2” give the “address” of the box, or cell.

Click the “Refresh” button and wait for the sheet to finish calculating.

A screenshot of the 'Prices' worksheet in Excel, showing a table of market data. The columns are: TypeID, StationID, Type, High, Low, Average, and a Refresh button. The data includes rows for items like 60001096 (Buy), 60003760 (Buy/Sell), 60004588 (Buy), 60005686 (Buy/Sell), 60006499 (Sell), 60008494 (Buy), 60011866 (Buy/Sell), and 60011866 (Sell).

TypeID	StationID	Type	High	Low	Average	
18	60001096	Buy	0.26	0.26	0.26	
18	60003760	Buy	37.30	21.01	31.89	
18	60003760	Sell	35,600.00	47.98	166.99	
18	60004588	Buy	28.12	27.05	27.34	
18	60005686	Buy	28.15	0.06	6.26	
18	60005686	Sell	37.00	28.16	36.56	
18	60006499	Sell	128.10	128.10	128.10	
18	60008494	Buy	29.70	5.02	21.78	
18	60011866	Buy	34.51	20.14	28.70	
18	60011866	Sell	39.69	34.53	37.14	

Customizing the Model

The intent of this document is to teach you how to model data from the Eve Online markets so some basic spreadsheet knowledge is required from this point onwards. Instructions will not be as detailed and there will be fewer step by step instructions.

Our focus will be on “how can I make this help me read the markets and not so much on how to do it procedurally.

Items Sheet – Watching more prices in the markets

The Items sheet can be used to track and verify prices for items in the WatchItems.csv file, a main configuration component of the Market Model.

This sheet can also be used to support hauling operations, basically buying items in one market, then transporting them and selling them in another for profit.

It can also be used to check market prices for items looted from NPC’s from ratting operations.

Since the data in the sheet is not a complete list of items in Eve, from time to time you may wish to add additional items. In this example, we will add a common item that drops from NPC's – the "5MN Y-T8 Compact Microwarpdrive."

You can find the information you need in the TypeIDs sheet.

647	648	649	650	651	652	653	654	655	656	657	658	659	660	661	662	663	664	665	666	667	668	669	670	671	672	673	674	675	676	677	678	679	680	681	682	683	684	685	686	687	688	689	690	691	692	693	694	695	696	697	698	699	700	701	702	703	704	705	706	707	708	709	710	711	712	713	714	715	716	717	718	719	720	721	722	723	724	725	726	727	728	729	730	731	732	733	734	735	736	737	738	739	740	741	742	743	744	745	746	747	748	749	750	751	752	753	754	755	756	757	758	759	760	761	762	763	764	765	766	767	768	769	770	771	772	773	774	775	776	777	778	779	780	781	782	783	784	785	786	787	788	789	790	791	792	793	794	795	796	797	798	799	800	801	802	803	804	805	806	807	808	809	810	811	812	813	814	815	816	817	818	819	820	821	822	823	824	825	826	827	828	829	830	831	832	833	834	835	836	837	838	839	840	841	842	843	844	845	846	847	848	849	850	851	852	853	854	855	856	857	858	859	860	861	862	863	864	865	866	867	868	869	870	871	872	873	874	875	876	877	878	879	880	881	882	883	884	885	886	887	888	889	890	891	892	893	894	895	896	897	898	899	900	901	902	903	904	905	906	907	908	909	910	911	912	913	914	915	916	917	918	919	920	921	922	923	924	925	926	927	928	929	930	931	932	933	934	935	936	937	938	939	940	941	942	943	944	945	946	947	948	949	950	951	952	953	954	955	956	957	958	959	960	961	962	963	964	965	966	967	968	969	970	971	972	973	974	975	976	977	978	979	980	981	982	983	984	985	986	987	988	989	990	991	992	993	994	995	996	997	998	999	1000	1001	1002	1003	1004	1005	1006	1007	1008	1009	1010	1011	1012	1013	1014	1015	1016	1017	1018	1019	1020	1021	1022	1023	1024	1025	1026	1027	1028	1029	1030	1031	1032	1033	1034	1035	1036	1037	1038	1039	1040	1041	1042	1043	1044	1045	1046	1047	1048	1049	1050	1051	1052	1053	1054	1055	1056	1057	1058	1059	1060	1061	1062	1063	1064	1065	1066	1067	1068	1069	1070	1071	1072	1073	1074	1075	1076	1077	1078	1079	1080	1081	1082	1083	1084	1085	1086	1087	1088	1089	1090	1091	1092	1093	1094	1095	1096	1097	1098	1099	1100	1101	1102	1103	1104	1105	1106	1107	1108	1109	1110	1111	1112	1113	1114	1115	1116	1117	1118	1119	1120	1121	1122	1123	1124	1125	1126	1127	1128	1129	1130	1131	1132	1133	1134	1135	1136	1137	1138	1139	1140	1141	1142	1143	1144	1145	1146	1147	1148	1149	1150	1151	1152	1153	1154	1155	1156	1157	1158	1159	1160	1161	1162	1163	1164	1165	1166	1167	1168	1169	1170	1171	1172	1173	1174	1175	1176	1177	1178	1179	1180	1181	1182	1183	1184	1185	1186	1187	1188	1189	1190	1191	1192	1193	1194	1195	1196	1197	1198	1199	1200	1201	1202	1203	1204	1205	1206	1207	1208	1209	1210	1211	1212	1213	1214	1215	1216	1217	1218	1219	1220	1221	1222	1223	1224	1225	1226	1227	1228	1229	1230	1231	1232	1233	1234	1235	1236	1237	1238	1239	1240	1241	1242	1243	1244	1245	1246	1247	1248	1249	1250	1251	1252	1253	1254	1255	1256	1257	1258	1259	1260	1261	1262	1263	1264	1265	1266	1267	1268	1269	1270	1271	1272	1273	1274	1275	1276	1277	1278	1279	1280	1281	1282	1283	1284	1285	1286	1287	1288	1289	1290	1291	1292	1293	1294	1295	1296	1297	1298	1299	1300	1301	1302	1303	1304	1305	1306	1307	1308	1309	1310	1311	1312	1313	1314	1315	1316	1317	1318	1319	1320	1321	1322	1323	1324	1325	1326	1327	1328	1329	1330	1331	1332	1333	1334	1335	1336	1337	1338	1339	1340	1341	1342	1343	1344	1345	1346	1347	1348	1349	1350	1351	1352	1353	1354	1355	1356	1357	1358	1359	1360	1361	1362	1363	1364	1365	1366	1367	1368	1369	1370	1371	1372	1373	1374	1375	1376	1377	1378	1379	1380	1381	1382	1383	1384	1385	1386	1387	1388	1389	1390	1391	1392	1393	1394	1395	1396	1397	1398	1399	1400	1401	1402	1403	1404	1405	1406	1407	1408	1409	1410	1411	1412	1413	1414	1415	1416	1417	1418	1419	1420	1421	1422	1423	1424	1425	1426	1427	1428	1429	1430	1431	1432	1433	1434	1435	1436	1437	1438	1439	1440	1441	1442	1443	1444	1445	1446	1447	1448	1449	1450	1451	1452	1453	1454	1455	1456	1457	1458	1459	1460	1461	1462	1463	1464	1465	1466	1467	1468	1469	1470	1471	1472	1473	1474	1475	1476	1477	1478	1479	1480	1481	1482	1483	1484	1485	1486	1487	1488	1489	1490	1491	1492	1493	1494	1495	1496	1497	1498	1499	1500	1501	1502	1503	1504	1505	1506	1507	1508	1509	1510	1511	1512	1513	1514	1515	1516	1517	1518	1519	1520	1521	1522	1523	1524	1525	1526	1527	1528	1529	1530	1531	1532	1533	1534	1535	1536	1537	1538	1539	1540	1541	1542	1543	1544	1545	1546	1547	1548	1549	1550	1551	1552	1553	1554	1555	1556	1557	1558	1559	1560	1561	1562	1563	1564	1565	1566	1567	1568	1569	1570	1571	1572	1573	1574	1575	1576	1577	1578	1579	1580	1581	1582	1583	1584	1585	1586	1587	1588	1589	1590	1591	1592	1593	1594	1595	1596	1597	1598	1599	1600	1601	1602	1603	1604	1605	1606	1607	1608	1609	1610	1611	1612	1613	1614	1615	1616	1617	1618	1619	1620	1621	1622	1623	1624	1625	1626	1627	1628	1629	1630	1631	1632	1633	1634	1635	1636	1637	1638	1639	1640	1641	1642	1643	1644	1645	1646	1647	1648	1649	1650	1651	1652	1653	1654	1655	1656	1657	1658	1659	1660	1661	1662	1663	1664	1665	1666	1667	1668	1669	1670	1671	1672	1673	1674	1675	1676	1677	1678	1679	1680	1681	1682	1683	1684	1685	1686	1687	1688	1689	1690	1691	1692	1693	1694	1695	1696	1697	1698	1699	1700	1701	1702	1703	1704	1705	1706	1707	1708	1709	1710	1711	1712	1713	1714	1715	1716	171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Finally, we need to add the formulas to look up the item prices from the Prices sheet. Select the price data from the row above (range E226:H226), right click on the selection and choose “Copy.”

	A	B	C	D	E	F	G	H
	ItemID	Item Name	Class		Jita Buy	Jita Sell	DDX Buy	
224	626	Vexor	Ship		9,990,000.00	11,990,000.00	9,550,000.00	
225	33003	Enormous Freight Container	Structure		238,400.00	397,400.00	500.00	
226	24445	Giant Freight Container	Structure		133,800.00	277,800.00	50,590.00	
227	5973	5MN Y-T8 Compact Microwarpdrive	Med Power					

Now, right-click in cell E227 and choose “Paste.”

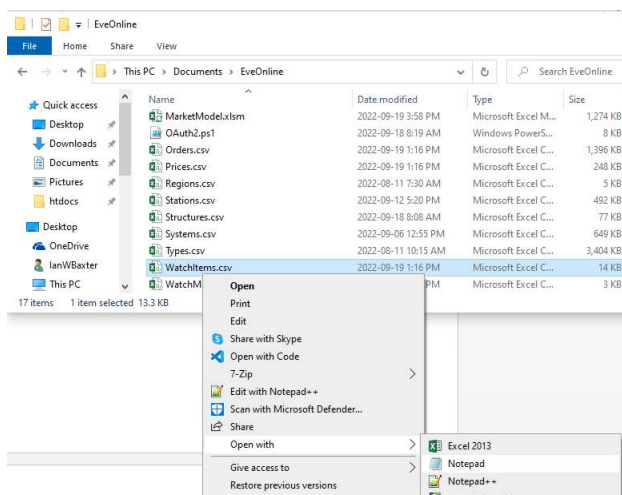
	A	B	C	D	E	F	G	H
	ItemID	Item Name	Class		Jita Buy	Jita Sell	DDX Buy	DDX Sell
224	626	Vexor	Ship					
225	33003	Enormous Freight Container	Structure					
226	24445	Giant Freight Container	Structure					
227	5973	5MN Y-T8 Compact Microwarpdrive	Med Power					

The data will update and you will now see “No Price” displayed in in the market price data.

	A	B	C	D	E	F	G	H
	ItemID	Item Name	Class		Jita Buy	Jita Sell	DDX Buy	DDX Sell
224	626	Vexor	Ship		9,990,000.00	11,990,000.00	9,550,000.00	11,250,000.00
225	33003	Enormous Freight Container	Structure		238,400.00	397,400.00	500.00	239,000.00
226	24445	Giant Freight Container	Structure		133,800.00	277,800.00	50,590.00	512,000.00
227	5973	5MN Y-T8 Compact Microwarpdrive	Med Power		No Price	No Price	No Price	No Price

This is because the WatchItems.csv file needs to be configured to collect the price data for that item. This is an easy thing to do. First, open the WatchItems.csv file in a text editor like Notepad.

To do this, use Windows Explorer, find and right-click on the file. Choose “Open with...” and then Notepad.



In Notepad, hit the key combination Ctrl-End to go to the end of the file. Switch to the Market Model and copy the data from the Items sheet columns A and B (the ID and name of the item). Paste it at the end of the file data in Notepad.

```
62530,"Compressed Rich Plagioclase"
5973    5MN Y-T8 Compact Microwarpdrive
```

The blank space between the ID and the name is a tab character, change that to a comma and make sure you also put double quotes around the item name.

```
62529,"Compressed Azure Plagioclase"
62530,"Compressed Rich Plagioclase"
5973,"5MN Y-T8 Compact Microwarpdrive"
```

Save the notepad file and close Notepad.

You now need to update the prices data using the Powershell script and then import the prices.csv data into the model. Once this is done, you should see prices listed in the Items sheet.

	A	B	C	D	E	F	G	H
1	ItemID	Item Name	Class		Jita Buy	Jita Sell	DDX Buy	DDX Sell
224	626	Vexor	Ship		9,987,000.00	11,970,000.00	9,552,000.00	12,100,000.00
225	33003	Enormous Freight Container	Structure		238,500.00	496,000.00	500.00	239,000.00
226	24445	Giant Freight Container	Structure		133,700.00	274,500.00	50,590.00	512,000.00
227	5973	5MN Y-T8 Compact Microwarpdrive	Med Power		261,000.00	298,500.00	155,000.00	197,200.00
228								

Hint: You can “organize” this sheet by using the spreadsheet application to sort the data. It is recommended to sort by “Class” first and then by “Item Name.”

Sort
?
X

Add Level
Delete Level
Copy Level
Options...
☒ My data has headers

Column	Sort On	Order
Sort by Class	Values	A to Z
Then by Item Name	Values	A to Z

OK
Cancel

Markets Sheet – Watching more markets

The Market Model will only watch prices in markets you specify. You can add or remove markets and have multiple markets in the same region. The regional market data is only queried once during the order download process in Powershell.

In this example, we will add a new market and configure it in Watchmarkets.csv

Switch to the “Markets” sheet.

106	Verge Vendor	10000068							
107	VR-01	14000001							
◀ ▶ Items Markets Watches Ore Salvage Planet T1 T2		Reg							
READY									

Enter the station name at the end of the data in column "C."

C18				
Pucherie VI - Moon 11 - Federation Navy Testing Facilities				
A		B		C
ID	Name	Station Name	Region Name	Region ID
17	60001096	Tash-Murkon	Tash-Murkon	10000020
18		Pucherie VI - Moon 11 - Federation Navy Testing Facilities		
19				

Switch to the RegionIDs sheet.

A screenshot of the Microsoft Excel ribbon. The 'RegionIDs' tab is highlighted in yellow. Other visible tabs include 'Items', 'Markets', 'Watches', 'Ore', 'Salvage', 'Planet', 'T1', 'T2', 'TypeIDs', 'Prices', and a plus icon for more tabs. The 'READY' status bar is visible at the bottom left of the ribbon area.

Find the entry for Sing Liason and copy the region name and ID.

	Region	ID
1	Scalding Pass	1000
90	Sinq Laison	1000
91	Solitude	1000
92	Stain	1000
93	Syndicate	1000
94	Tash-Murkon	1000
95	Tenal	1000

Paste in in the Markets sheet in the cell for the Region name.

D18				
	A	B	C	D
1	ID	Name	Station Name	Region ID
17	60001096	Tash-Murkon	Tash-Murkon Prime II - Moon 11 - Kaalakiota Corporation Factor	10000020
18			Pucherie VI - Moon 11 - Federation Navy Testing Facilities	10000032

Enter Pucherie in the “Name” column. This is an “alias” we will use for that station.

You can find the Station ID (we call it Market ID) in the stations.csv file. It is the first item in the line. Note that NPC stations use ID's starting with 6000000 where player structures (in structures.csv) start with 10000000000000.

Paste the Station ID in column “A” of the Markets sheet.

A12					
1032903563769					
	A	B	C	D	E
	ID	Name	Station Name	Region Name	Region ID
17	60001096	Tash-Murkon	Tash-Murkon Prime II - Moon 1 - Kaalakiota Corporation Factor	Tash-Murkon	10000020
18	60011857	Pucherie	Pucherie VI - Moon 11 - Federation Navy Testing Facilities	Sinq Laison	10000032

Now we have to add the market to the WatchMarkets.csv file. Open it in Notepad (or your favorite text editor) and hit Ctrl-End to go to the last line. Switch back to the Markets sheet and copy the data for the market and paste it to the end of WatchMarkets.csv

```
*WatchMarkets.csv - Notepad
File Edit Format View Help
60001096,"Tash-Murkon","Tash-Murkon Prime II - Moon 1 - Kaalakiota Corporation Factory","Tash-Murkon",10000020
60006499,"Teonusude","Teonusude III - Moon 8 - Imperial Armaments Factory","Molden Heath",10000028
1032792618788,"Villore","Villore - IChooseYou Market and Industry","Essence",10000064
60011857      Pucherie      Pucherie VI - Moon 11 - Federation Navy Testing Facilities      Sinq Laison      10000032
```

Replace the tabs with commas and make sure that the text data is enclosed in double quotes. Save the file when you are done.

```
WatchMarkets.csv - Notepad
File Edit Format View Help
1032714784769,"Sobaseki","Sobaseki - IChooseYou Market and Industry","LoneTrek",10000016
60001096,"Tash-Murkon","Tash-Murkon Prime II - Moon 1 - Kaalakiota Corporation Factory","Tash-Murkon",10000020
60006499,"Teonusude","Teonusude III - Moon 8 - Imperial Armaments Factory","Molden Heath",10000028
1032792618788,"Villore","Villore - IChooseYou Market and Industry","Essence",10000064
60011857,"Pucherie","Pucherie VI - Moon 11 - Federation Navy Testing Facilities","Sinq Laison",10000032
```

Run the public data collection script again and update the prices data in the model. You will need it for the next section.

Items sheet – “Hauling” model (Adding markets, math, conditional formatting)

In addition to providing a backup for the WatchItems.csv file, the Items data can be used to identify “market opportunities” for buying items in one market and then re-selling them in another at a profit.

The limitation of the model is that it is not real-time data and by the time you get to the source or destination, the opportunity could have been lost. Market browsers such as [Eve Tycoon](#) also have this limitation since the data is not real-time. Nothing is guaranteed in Eve, but the model is better than guessing.

The Items sheet is already set up for Dodixie and Jita, listing Buy and Sell order prices for each. If you plan to sell immediately, you would be selling items to fill a Buy order with the highest price in that case. If you choose to place items for sale over time, then you would be looking at the lowest Sell order price.

B19

125mm Railgun II

	A	B	C	D	E	F	G	H	I
1	ItemID	Item Name	Class		Jita Buy	Jita Sell		DDX Buy	DDX Sell
2	230	Antimatter Charge M	Charge		59.07	66.90		35.01	58.97
3	225	Iridium Charge M	Charge		32.02	59.77		6.00	45.50
4	237	Plutonium Charge L	Charge		49.54	115.80		No Price	138.00
5	229	Plutonium Charge M	Charge		45.26	72.94		45.30	79.90
6	221	Plutonium Charge S	Charge		11.30	18.98		6.86	19.20
7	224	Tungsten Charge M	Charge		24.07	29.45		1.44	38.49
8	228	Uranium Charge M	Charge		51.00	64.00		13.13	54.99

When you sell items on the market, you also need to pay a broker’s fee and taxes. Broker fees can be reduced by training the Broker Relations skill, and taxes can be reduced by training the Accounting skill. As a general rule, broker fees and taxes are usually around 10%, but can be reduced to 6.6% by fully

training those skills. This means that you must make about 10% profit to break even, and more than 10% to actually profit.

Profit percentage can be calculated as $((\text{Sell Price} - \text{Taxes and Fees}) / \text{Buy Price}) - 1 \times 100$

In the example above, it looks like there is an opportunity buying “Antimatter Charge M” in Dodixie for 58.97 ISK each and then selling in Jita over time at 66.75 ISK (just below the lowest Sell order at 66.90 ISK). Taking into account for taxes and fees, you would get 60.075 ISK ($66.75 \times .9$) selling at 66.75 ISK.

The formula then would be $((66.75 \times .9) / 58.97) - 1 \times 100$, or 1.87% profit. With the effort involved and the risk of being undercut selling over time in Jita, it seems hardly worth the time, yet people do it and lose money.

Using this example, we will add another market to the Items sheet. There is always a greater profit potential selling items in non-hub markets. The market we just added in the last section, Pucherie, is just such a market.

Select the entire columns “H” and “I”, right-click on them and select “Copy” from the dropdown menu.

ItemID	Item Name	Class	Jita Buy	Jita Sell	DDX Buy	DDX Sell
230	Antimatter Charge M	Charge	59.07	66.90	35.01	58.97
225	Iridium Charge M	Charge	32.02	59.77	6.00	45.50
237	Plutonium Charge L	Charge	49.54	115.80	No Price	138.00
229	Plutonium Charge M	Charge	45.26	72.94	45.30	79.90
221	Plutonium Charge S	Charge	11.30	18.98	6.86	19.20
224	Tungsten Charge M	Charge	24.07	29.45	1.44	38.49
228	Uranium Charge M	Charge	51.00	64.00	13.13	54.99
2454	Hobgoblin I	Drone	3,017.00	3,328.00	1,854.00	4,000.00

Select Cell “K1”, right-click in it and paste the data you just copied. The sheet will recalculate.

ItemID	Item Name	Class	Jita Buy	Jita Sell	DDX Buy	DDX Sell	Pucherie Buy	Pucherie Sell
230	Antimatter Charge M	Charge	59.07	66.90	35.01	58.97	35.01	58.97
225	Iridium Charge M	Charge	32.02	59.77	6.00	45.50	6.00	45.50
237	Plutonium Charge L	Charge	49.54	115.80	No Price	138.00	No Price	138.00
229	Plutonium Charge M	Charge	45.26	72.94	45.30	79.90	45.30	79.90
221	Plutonium Charge S	Charge	11.30	18.98	6.86	19.20	6.86	19.20
224	Tungsten Charge M	Charge	24.07	29.45	1.44	38.49	1.44	38.49
228	Uranium Charge M	Charge	51.00	64.00	13.13	54.99	13.13	54.99
2454	Hobgoblin I	Drone	3,017.00	3,328.00	1,854.00	4,000.00	1,854.00	4,000.00
10246	Mining Drone I	Drone	7,399.00	9,988.00	5,001.00	6,980.00	5,001.00	6,980.00

In cells “K1” and “L1” change “DDX” to “Pucherie”. After the recalculation, you now have Pucherie prices listed.

ItemID	Item Name	Class	Jita Buy	Jita Sell	DDX Buy	DDX Sell	Pucherie Buy	Pucherie Sell
230	Antimatter Charge M	Charge	59.07	66.90	35.01	58.97	No Price	No Price
225	Iridium Charge M	Charge	32.02	59.77	6.00	45.50	No Price	No Price
237	Plutonium Charge L	Charge	49.54	115.80	No Price	138.00	No Price	No Price
229	Plutonium Charge M	Charge	45.26	72.94	45.30	79.90	No Price	No Price
221	Plutonium Charge S	Charge	11.30	18.98	6.86	19.20	No Price	No Price
224	Tungsten Charge M	Charge	24.07	29.45	1.44	38.49	No Price	No Price

Notice the formula in the formula bar, this is really what is doing all the work. If you click inside this formula, the spreadsheet will highlight the cells it uses for easier identification.

ItemID	Item Name	Class	Jita Buy	Jita Sell	DDX Buy	DDX Sell	Pucherie Buy	Pucherie Sell
230	Antimatter Charge M	Charge	59.07	66.90	35.01	58.97	No Price	=Get_Market_Pr
225	Iridium Charge M	Charge	32.02	59.77	6.00	45.50	No Price	No Price
237	Plutonium Charge L	Charge	49.54	115.80	No Price	138.00	No Price	No Price
229	Plutonium Charge M	Charge	45.26	72.94	45.30	79.90	No Price	No Price

How this works:

=Get_Market_Price(\$A2,Get_MarketID(Get_MarketName(L\$1)),"Buy","High")

1. The market name, "Pucherie" is collected from cell L\$1 with the Get_Market_Name custom function.
2. The market ID is collected from the market name "Pucherie" with the Get_Market_ID custom function.
3. The value in cell "\$A2" is substituted for the cell address
4. The custom function call Get_Market_Price then looks like this:
=Get_Market_Price(230, 60011857,"Buy","High")
5. The Get_Market_Price custom function looks up the data for the item, market id and order type on the Prices sheet to return the high price back to the sheet.

Order types that can be specified are: "Buy" or "Sell." Price data can be "High", "Low" or "Average."

Pucherie doesn't seem to be a very active market. Remove it from the Items sheet by selecting columns "K" and "L", right-clicking on the selection and choosing "Delete" from the dropdown menu.

Also delete the row in the Markets sheet for Pucherie, and the line holding Pucherie data from Watchmarkets.csv to save time downloading the prices data.

A profit calculation can be added to the Items sheet to make it easier to identify profit opportunities. In this case, we will see if we can buy stuff in Jita and sell it over time for profit in Dodixie. Select cell "K1" and enter Jita->DDX so you know what data we will display in that column.

Item#	Item Name	Class	Jita Buy	Jita Sell	DDX Buy	DDX Sell	Jita->DDX
230	Antimatter Charge M	Charge	59.07	66.90	35.01	58.97	
225	Iridium Charge M	Charge	32.02	59.77	6.00	45.50	
237	Plutonium Charge L	Charge	49.54	115.80	No Price	138.00	
229	Plutonium Charge M	Charge	45.26	72.94	45.30	79.90	

In cell "K2" enter the following formula: $=((\$I2*0.9)/\$F2)-1$ (without the quotes)

Note that we have a dollar sign before the column portion of the cell address. This tells the spreadsheet that if the formula is pasted into another cell that this "column" part of the cell address should not change. You can do this with the row component of the address (ie. A\$2), or the entire cell address (ie. \$A\$2.)

Right-Click on cell "K2" and select "Format cells" from the dropdown menu.

DDX Sell	Jita->DDX
35.01	58.97
6.00	45.50
Price	138.00
45.30	79.90
6.86	19.20
1.44	38.49
13.13	54.99
54.00	4,000.00
01.00	6,980.00
00.00	623,500.00
70.00	59,910.00
10.00	56,990.00
51.00	6,000.00
90.00	43,440.00
00.00	4,389,000.00
00.00	3,249,000.00
50.00	24,860.00
00.00	1,261,000.00
10.00	34,900.00
00.00	848,000.00
00.00	740,000.00

Format Cells

Number Alignment Font Border Fill Protection

Category:

- General
- Number
- Currency
- Accounting
- Date
- Time
- Percentage**
- Fraction
- Scientific
- Text
- Special
- Custom

Sample
2.10%

Decimal places: 2

[illegible]

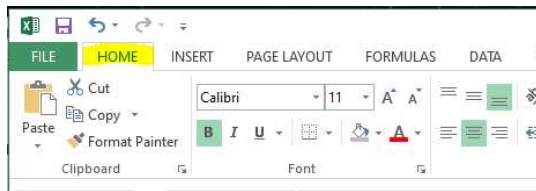
Some cells with show #VALUE. This indicates that one of the prices needed for the profit calculation is missing. No order of the type we are looking for exists on the market if it says “No Price.”

If you want to make a profit using immediate sales, you need to change the formula based on what you have learned. Change the formula in cell "K2" to " $=((\$H2*0.9)/\$F2)-1$ " and copy it down to all rows with item data.

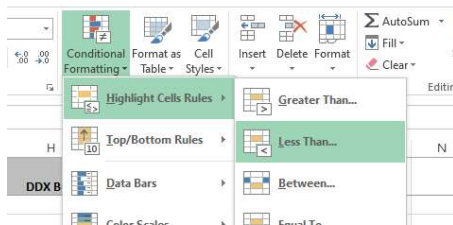
To make it easier to quickly identify what is profitable and what is not, you can use conditional formatting to color losses as red. Use the instructions following for the spreadsheet product you are using.

Conditional formatting – Excel

Make sure you have the “Home” tab selected on the ribbon bar.



Select the entire column “K” and then click the Conditional Formatting icon on the ribbon bar and choose “Highlight Cells Rules”, “Less Than...”



Use the options below:

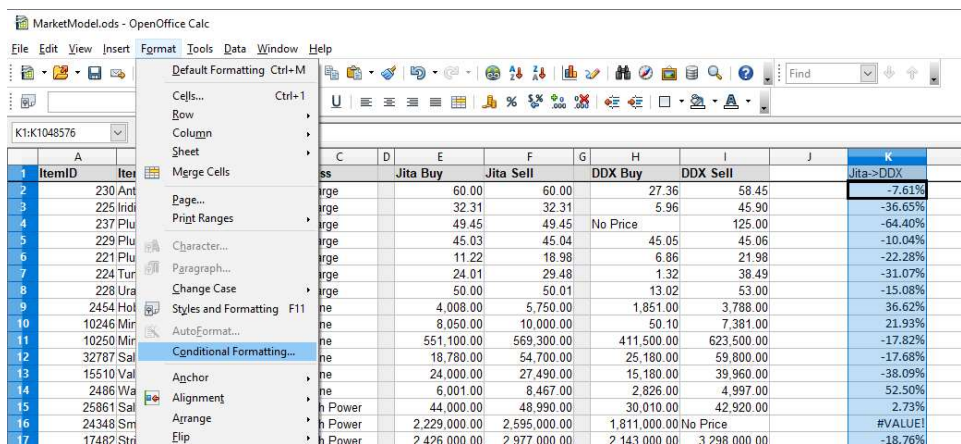


Any loss will now show as red.

	A	B	C	D	E	F	G	H	I	J	K
1	ItemID	Item Name	Class	Jita Buy	Jita Sell	DDX Buy	DDX Sell	Jita->DDX			
2	230	Antimatter Charge M	Charge	59.07	66.90	35.01	58.97	-20.67%			
3	225	Iridium Charge M	Charge	32.02	59.77	6.00	45.50	-31.49%			
4	237	Plutonium Charge L	Charge	49.54	115.80	No Price	138.00	7.25%			
5	229	Plutonium Charge M	Charge	45.26	72.94	45.30	79.90	-1.41%			
6	221	Plutonium Charge S	Charge	11.30	18.98	6.86	19.20	-8.96%			

Conditional formatting - OpenOffice

Select the entire column “K” and from the menu select “Format”, “Conditional Formatting...”



Select the options below:

Conditional Formatting

☒ Condition 1

Cell value is

Cell Style

☐ Condition 2

Cell value is

Any loss will now show as red.

1	ItemID	Item Name	Class	Jita Buy	Jita Sell	DDX Buy	DDX Sell	Jita->DDX
2	230	Antimatter Charge M	Charge	60.00	60.00	27.36	58.45	-7.61%
3	225	Iridium Charge M	Charge	32.31	32.31	5.96	45.90	-36.65%
4	237	Plutonium Charge L	Charge	49.45	49.45	No Price	125.00	-64.40%
5	229	Plutonium Charge M	Charge	45.03	45.04	45.05	45.06	-10.04%
6	221	Plutonium Charge S	Charge	11.22	18.98	6.86	21.98	-22.28%
7	224	Tungsten Charge M	Charge	24.01	29.48	1.32	38.49	-31.07%
8	228	Uranium Charge M	Charge	50.00	50.01	13.02	53.00	-15.08%
9	2454	Hobgoblin I	Drone	4,008.00	5,750.00	1,851.00	3,788.00	36.62%
10	10246	Mining Drone I	Drone	8,050.00	10,000.00	50.10	7,381.00	21.93%

Ore sheet – “Miner” model (Adding ore and minerals)

The Ore sheet is designed to answer the question of “how do I make the most from what I mine?”

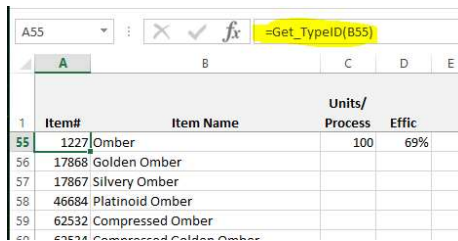
It has 5 parts:

- Information (Green) – Item ID, Name and data needed for reprocessing calculations.
- Market Buy/Sell (Blue) – What is the item selling and being bought for in the market?
- Compressed Markey Buy/Sell (Purple) – If it can be compressed, what are the compressed item market buy and sell prices?
- Refined Market Buy/Sell (Yellow) – If I refine it to minerals, what is their combined buy or sell price?
- Refined results (Red) – Data required for the refined market price, if I refine it, what does it make?

Item#	Item Name	Units/Process	Effic	Jita Buy	Jita Sell	Jita Buy Compressed	Jita Sell Compressed	Jita Buy Minerals	Jita Sell Minerals	Titanium	Mexallon	Pyerite	Oxygen Isotopes	Helium Isotopes	Hydroge Isotope
18	Plagioclase	100	69%	38.53	43.00	40.50	43.00	38.63	40.98	175	70				
17455	Azure Plagioclase	100	69%	37.57	37.58	46.68	47.00	40.81	43.29	184	74				
17456	Rich Plagioclase	100	69%	40.23	45.00	48.26	50.99	42.51	45.09	193	77				
1228	Scordite	100	69%	14.38	17.00	17.26	22.00	9.10	9.64	150		90			
17463	Condensed Scordite	100	69%	14.38	15.25	15.71	17.70	9.60	10.17	158		95			
17464	Massive Scordite	100	69%	13.25	17.75	17.60	22.00	10.01	10.60	165		99			
1230	Veldspar	100	69%	9.96	12.50	11.87	13.55	10.46	10.98	400					
17470	Concentrated Veldspar	100	69%	10.54	10.63	12.32	13.75	10.98	11.53	420					
17471	Dense Veldspar	100	69%	11.15	11.16	13.89	14.74	11.51	12.08	440					
16264	Blue Ice	1	69%	1,401.00	152,000.00	136,300.00	147,400.00	132,833.66	147,022.23					414	
28433	Compressed Blue Ice	1	69%	136,300.00	147,400.00	---	---	132,833.66	147,022.23					414	

There will be cases where, say for ice or low sec ore, they need to be added to the model. For this example, we will use Omber. There’s 4 types of Omber, and then there are the compressed versions to add. If you have some old compressed ore, you will also need to add the batch compressed versions of the ore as well.

You can add the formula “=Get_TypeID()” to verify that everything in the item name is spelled correctly. If it isn’t, the Type ID will return a -1 instead of the correct ID.



	A	B	C	D	E
1	Item#	Item Name	Units/ Process	Effic	
55	1227	Omber	100	69%	
56	17868	Golden Omber			
57	17867	Silvery Omber			
58	46684	Platinoid Omber			
59	62532	Compressed Omber			
60	62531	Compressed Golden Omber			

Using the instructions provided earlier, add the ores to the Items sheet and the WatchItems.csv.

You can select the ore in the regional market, right-click on it to “Show Info” and click the “Attributes” tab. You can then find the value for the “Units/Process” column.



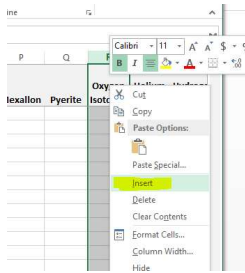
The value for the reprocessing efficiency can be obtained when you are in your favorite refinery. Right-click on some ore and select “Reprocess” and the efficiency will be shown in the dialog.



In this case, Omber refines into Pyerite and Isogen. Isogen is not listed in the Ore sheet, so it needs to be added to the “refined results” section. It doesn’t need to be added elsewhere, it’s already in the Items sheet.



To make it really easy, just insert a column after the “Pyerite” column. This way the formulas will “adjust” to include the inserted column.



Enter how many Pyerite and Isogen result from the refining.

Item#	Item Name	Units/Process	Effic	Jita Buy	Jita Sell	Jita Buy Compressed	Jita Sell Compressed	Jita Buy Minerals	Jita Sell Minerals	Tritanium	Mexallon	Pyerite	Isogen	Oxy Isc
1227	Omber	100	69%										90	75
17868	Golden Omber													
17867	Silvery Omber													

Repeat this for each of the ores listed.

Make sure that the formula for the “Sell Minerals” has all the result columns selected, in this case outlined in blue. If not, adjust the formulas for the “Buy Minerals” and “Sell Minerals” columns and make sure they are all updated.

Jita Sell Minerals	Tritanium	Mexallon	Pyerite	Isogen	Oxygen Isotopes	Helium Isotopes	Hydrogen Isotopes	Nitrogen Isotopes	Liquid Ozone	Heavy Water	Strontium Clathrates	Hydrocarbons	Silicates	Evaporite Deposits	Atmospheric Gases
2,210.57		800	4000										130		
2,290.88		800	8000											130	
["", "Low"]/\$C5		800	16000												130
				90	75										
				99	83										
				95	79										
				104	86										

Copy and paste the formulas down in each column.

Item#	Item Name	Units/Process	Effic	Jita Buy	Jita Sell	Jita Buy Compressed	Jita Sell Compressed	Jita Buy Minerals	Jita Sell Minerals	Tritanium	Mexallon	Pyerite	Isogen	Oxy Isc
1227	Omber	100	69%	No Price	No Price	No Price	No Price	170.83	197.77				90	75
17868	Golden Omber	100	69%	No Price	No Price	No Price	No Price	189.02	218.83				99	83
17867	Silvery Omber	100	69%	No Price	No Price	No Price	No Price	179.95	208.33				95	79
46684	Platinoid Omber	100	69%	No Price	No Price	No Price	No Price	155.93	226.83				104	86
62532	Compressed Omber	100	69%	No Price	No Price	---	---	170.83	197.77				90	75
62534	Compressed Golden Omber	100	69%	No Price	No Price	---	---	189.02	218.83				99	83
62533	Compressed Silvery Omber	100	69%	No Price	No Price	---	---	179.95	208.33				95	79
62535	Compressed Platinoid Omber	100	69%	No Price	No Price	---	---	155.93	226.83				104	86
28416	Batch Compressed Omber	1	69%	No Price	No Price	---	---	17,083.09	19,777.19				90	75
28415	Batch Compressed Golden Omber	1	69%	No Price	No Price	---	---	18,901.83	21,883.08				99	83
28417	Batch Compressed Silvery Omber	1	69%	No Price	No Price	---	---	17,995.34	20,833.20				95	79
46700	Batch Compressed Platinoid Omber	1	69%	No Price	No Price	---	---	19,593.21	22,682.76				104	86

Update the prices following the documentation.

Item#	Item Name	Units/Process	Effic	Jita Buy	Jita Sell	Jita Buy Compressed	Jita Sell Compressed	Jita Buy Minerals	Jita Sell Minerals	Tritanium	Mexallon	Pyerite	Isogen	Oxy Isc
1227	Omber	100	69%	280.50	369.00			270.00	399.00			198.69	209.03	90 75
17868	Golden Omber	100	69%	250.00	299.90			300.00	385.00			219.85	231.29	99 83
17867	Silvery Omber	100	69%	200.50	232.50			271.20	399.00			209.30	220.19	95 79
46684	Platinoid Omber	100	69%	109.10	No Price			2,000.00	121,200.00			227.88	239.73	104 86
62532	Compressed Omber	100	69%	270.00	399.00	---	---					198.69	209.03	90 75
62534	Compressed Golden Omber	100	69%	300.00	385.00	---	---					219.85	231.29	99 83
62533	Compressed Silvery Omber	100	69%	271.20	399.00	---	---					209.30	220.19	95 79
62535	Compressed Platinoid Omber	100	69%	2,000.00	121,200.00	---	---					227.88	239.73	104 86
28416	Batch Compressed Omber	1	69%	22,130.00	43,000.00	---	---					19,868.90	20,902.86	90 75
28415	Batch Compressed Golden Omber	1	69%	21,100.00	No Price	---	---					21,984.75	23,128.83	99 83
28417	Batch Compressed Silvery Omber	1	69%	1,701.00	No Price	---	---					20,929.74	22,018.90	95 79
46700	Batch Compressed Platinoid Omber	1	69%	801.20	No Price	---	---					22,787.66	23,973.50	104 86

T1 sheet – “Maker” model (Adding blueprints and materials)

T2 sheet – “Maker” model (Adding blueprints and materials)

Custom Functions

Get_Market_Price

Get_Materials_Price

Get_TypeID

Get_MarketID

Get_MarketName

Get_RegionID

Get_MakePrice